

Employment

Graduate Researcher, IRL Lab	University of Alberta	May 2024 – Aug 2026
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- Built **Multi-Stew**, a three-service Overcooked platform for human-AI teaming research with Aggregate Intellect Inc.: Colyseus game server, React frontend over WebSockets, and FastAPI RL server, deployed to a VPS with Docker Compose.
- Designed the RL server as one endpoint with training, inference, and XAI modes, so a single service both trains agents and serves them to live players.
- Experimented with multi-agent RL algorithms for human-AI team.

RL Researcher	PCL/University of Alberta	May 2024 – Aug 2024
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- Built a simulation environment for a modular construction task with industry partner PCL; trained DQN and PPO with Ray RLlib across parallel workers on an HPC cluster.
- Set up cached GitHub Actions CI for the Python codebase.

Teaching Assistant	University of Alberta	Sep 2022 – Aug 2026
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- Courses: Algorithms; Formal Systems and Logic for CS; Data Analytics and Visualization.

Education

Edmonton, AB	University of Alberta	September 2024 – August 2026
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- MSc Computing Science — Reinforcement Learning and human-AI teaming. Supervised by Matthew E. Taylor and Adam S. Parker.

Edmonton, AB	University of Alberta	September 2020 – April 2024
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- B.Sc. Computing Science with Honors GPA: 3.7
- Coursework: Compiler Design; Intermediate Machine Learning; Parallel and Distributed Systems; Operating Systems; Computer Architecture; Algorithms; Databases; Web Application and Architecture.

Projects

Gazprea Compiler	2023
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- Compiler for the Gazprea language (developed by IBM), written in C++ by a team of 4; ANTLRv4 frontend emitting MLIR-IR in the LLVM dialect.
- Built two AST passes for typechecking and symbol table management.
- Compiles advanced programs including depth-first search, Fibonacci, and 8-Queens.

Parallel Sorting by Regular Sampling (🔗)	2024
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- Implemented the PSRS parallel sorting algorithm in C and benchmarked scaling across up to 128 cores.

Robotics in Duckietown (link)	2025
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- PID-controlled wheeled robot deployed on real hardware.

Technical Skills

- **Languages:** C/C++, Python, Java, JavaScript/TypeScript, SQL, Assembly (RISC-V, x86, ARM)
- **ML & RL:** PyTorch, Ray RLlib, multi-agent RL, HPC/distributed training
- **Compilers & Systems:** MLIR, LLVM, ANTLRv4, concurrency, parallel programming
- **Web & Tools:** React, Node/Colyseus, FastAPI, Docker, Apptainer, Git, GitHub Actions CI, Linux